

SCHOOL OF CHEMICAL & MATERIALS ENGINEERING
NATIONAL UNIVERSITY OF SCIENCES & TECHNOLOGY, ISLAMABAD

CHE-226 Programming and Data Science

Assignment 1 • Python for Chemical Engineering Calculations

Topics:	T1–T5 (Python basics to lists & tuples)	Total marks:	40
Submission:	One Jupyter notebook + its PDF export	Due:	Friday 30 October 2026, 11:59 pm
Assesses:	CLO-1 at Bloom level C2 (understand): code is the tool, explanation earns the marks		

Instructions

- **Tools.** Use only what we covered in T1–T5: variables, arithmetic, f-strings, the math module, if/elif/else, for/while, functions (default and keyword arguments, lambda, map, filter), lists, tuples, slicing, sorted, comprehensions and zip. **Do not use NumPy, pandas or Matplotlib** in this assignment.
- **Notebook layout.** One section per problem. Each task (a), (b), ... gets its own code cell, run top to bottom with outputs visible. Copy the *Given* data exactly as printed.
- **Interpretation.** Every problem ends with a markdown cell headed *Interpretation* that answers the last task in full sentences, quoting your own numbers with units. Code without interpretation earns at most half the marks for that problem.
- **Style.** Meaningful variable names, a one-line docstring for every function, units in comments, and printed tables aligned with f-string widths (e.g. {P:>8.2f}).
- **Naming.** RegNo_A1.ipynb and RegNo_A1.pdf. Work individually; matching code between students is treated as copying.

No.	Chemical engineering context	Ontology topics	Python skills (lecture)	Marks
1	Real-gas pressure in a CO ₂ storage cylinder	T-02.01.01 Gas Laws and Real Gases; T-04.02.02 Equations of State	functions with defaults (T4), for/while (T3), tuples (T5)	10
2	Mixing BTX streams and checking plant data	T-03.02.01 Single-Unit Balances; T-03.01.02 Composition Measures	filter/lambda (T4), zip, comprehensions, sorted with key (T5)	10
3	Reaction rate vs temperature in a batch reactor	T-08.01.02 Arrhenius Behaviour; T-08.02.01 Batch Reactor; T-01.04.03 Numerical ODE Solution	math (T2), while (T3), slicing ks[1:] (T5)	10
4	Detecting fouling from heat-exchanger logs	T-06.04.01 Overall Heat Transfer Coefficient; T-06.04.02 LMTD Method; T-01.05.02 Error and Uncertainty Analysis	sentinel & continue (T3), tuple unpacking, min with key (T5)	10

Marking guide

Criterion	Full marks	Partial	Little or none
Code works (tasks a–c)	Runs top to bottom; correct numbers; asked-for constructs used	Minor errors, or the right numbers reached without the requested construct	Does not run, or wrong method
Output	Aligned tables, units, sensible decimal places	Correct but hard to read	Raw lists or no units
Interpretation (last task)	Answers every sub-part with your own numbers and the engineering reason	Restates numbers without explaining them, or misses a sub-part	Missing, generic, or contradicts the output

Problem 1 How far can you trust the ideal-gas law?

CLO-1 · C2 · 10 marks

Ontology: T-02.01.01 Gas Laws and Real Gases • T-04.02.02 Equations of State • T-03.01.01 Units and Dimensions

A 50.0 L steel cylinder holds 5.00 kg of CO₂. The cylinder sits outdoors, so its temperature can range from 280 K on a winter night to 360 K in direct summer sun. Its relief valve opens at 55 bar. The safety engineer wants to know how much the pressure estimate depends on the gas model used.

$$P_{\text{ideal}} = \frac{nRT}{V} \quad P_{\text{vdw}} = \frac{nRT}{V - nb} - \frac{a n^2}{V^2} \quad Z = \frac{P_{\text{vdw}}}{P_{\text{ideal}}}$$

Given

1	m = 5000.0	# g of CO ₂
2	M = 44.01	# g/mol
3	V = 50.0	# L
4	R = 0.08314	# L.bar/(mol.K)
5	a = 3.640	# L ² .bar/mol ² (van der Waals, CO ₂)
6	b = 0.04267	# L/mol

- (a) Write two functions, `p_ideal(n, V, T, R=0.08314)` and `p_vdw(n, V, T, a, b, R=0.08314)`, that return pressure in bar. **Estimate** the number of moles `n` and print it to 2 decimal places. [2]
- (b) For `T = 280, 300, ..., 360 K` (use range), build a list of tuples (`T, P_ideal, P_vdw, Z`). Then loop over the list and **summarize** it as an aligned table with a header row (pressures to 2 decimals, `Z` to 3 decimals). [3]
- (c) Using a while loop that starts at 280 K and steps by 1 K, **infer** the first temperature at which the pressure reaches the 55 bar relief setting. Do this once with each model and print both temperatures and the difference between them. [2]
- (d) **Interpret** your results in a markdown cell: (i) how does `Z` change with temperature, and what does `Z < 1` say about the molecules? (ii) which model is the *conservative* choice for sizing the relief valve, and why? (iii) would you trust the ideal-gas law for this cylinder? Support your answer with numbers. [3]

Problem 2 Mixing BTX streams: check the data before you balance CLO-1 · C2 · 10 marks

Ontology: T-03.02.01 Single-Unit Balances • T-03.01.02 Composition Measures • T-01.09.04 Data Interpretation

Four streams of benzene (B), toluene (T) and xylene (X) are combined in a mixing tank that runs at steady state with no reaction. Each record below holds the stream name, its molar flow in kmol/h, and its mole fractions $[x_B, x_T, x_X]$. One of the records has been typed in wrongly at the plant.

Given

```

1 | streams = [("S1", 120.0, [0.50, 0.30, 0.20]),
2 |           ("S2", 80.0, [0.20, 0.50, 0.30]),
3 |           ("S3", 50.0, [0.10, 0.20, 0.70]),
4 |           ("S4", 40.0, [0.40, 0.40, 0.28])]
5 | names = ["benzene", "toluene", "xylene"]
6 | M = [78.11, 92.14, 106.17] # g/mol, same order as names

```

- (a) Write `is_valid(x, tol=0.01)`, which returns True when the mole fractions in `x` sum to 1 within `tol`. Use `filter` with a `lambda` to keep only the valid streams, and **identify** the names of any that are rejected. [2]
- (b) Write `mix(stream_list)`, which returns a tuple (`component_flows`, `total_flow`, `y`), where `y` holds the product mole fractions. Use a list comprehension for the component balance $\dot{n}_i = \sum_j F_j x_{ij}$. Run it on the valid streams and **summarize** the results. [3]
- (c) **Translate** the product into mass flow (kg/h) and mass fractions using `zip(component_flows, M)`. **Classify** the components by mass fraction, largest first, using `sorted(..., key=...)`. [2]
- (d) **Interpret:** (i) **compare** the mole and mass compositions. Why does xylene's share grow on a mass basis? (ii) Run `mix` on *all four* streams. How far apart are the sum of the component flows and the total feed (kmol/h)? What would this do to a plant balance? (iii) In two sentences, **explain** why the validation in (a) must come before any balance is solved. [3]

Problem 3 Hotter is faster, but by how much?

CLO-1 · C2 · 10 marks

Ontology: T-08.01.02 Arrhenius Behaviour · T-08.02.01 Batch Reactor · T-03.03.02 Conversion · T-01.04.03 Numerical ODE Solution

A liquid-phase reaction $A \rightarrow B$ is first order and runs in an isothermal batch reactor with $C_{A0} = 2.0$ mol/L. Its rate constant follows the Arrhenius law, and the batch is finished at 90 % conversion.

$$k = A e^{-E_a/RT} \quad \frac{dC_A}{dt} = -k C_A \quad \Rightarrow \quad t_{90} = \frac{-\ln(1 - 0.9)}{k}$$

Given $A = 5.0 \times 10^{10} \text{ min}^{-1}$, $E_a = 75\,000 \text{ J/mol}$, $R = 8.314 \text{ J/(mol K)}$.

- Write `rate_constant(T, A=5.0e10, Ea=75000.0, R=8.314)`, returning k in min^{-1} . For T from 300 to 360 K in steps of 10 K, **estimate** k (scientific notation, e.g. $.3e$) and t_{90} in minutes. [2]
- Store the k values in a list `ks`. Using `zip(ks, ks[1:])` in a list comprehension, **compare** each k with the next one: print the factor by which k rises per 10 K step, to 2 decimals. [2]
- Write `simulate_batch(k, dt, X_target=0.9, CA0=2.0)`, which steps the balance forward in time (Euler method: $CA = CA - k*CA*dt$) inside a while loop until the conversion $X = (C_{A0} - C_A)/C_{A0}$ reaches X_{target} . It returns (t, CA) . At 340 K, run it with $dt = 2.0, 0.5$ and 0.1 min, and **compare** each time with the exact t_{90} (% error). [3]
- Interpret:** (i) does the textbook rule “the rate doubles every 10 K” hold here? How does the factor change with temperature? (ii) The plant can run one batch per shift (8 h). What is the lowest temperature in your table that meets this target? What might stop you from simply choosing 360 K? (iii) **Explain** why the Euler result changes with dt , and whether it finishes the batch too early or too late. [3]

Problem 4 Is the heat exchanger fouling?

CLO-1 · C2 · 10 marks

Ontology: T-06.04.01 Overall Heat Transfer Coefficient • T-06.04.02 LMTD Method • T-03.04.01 Enthalpy and Heat Capacity • T-01.05.02 Error and Uncertainty Analysis

A counter-current exchanger cools hot oil with cooling water. Operators logged the four terminal temperatures (°C) every hour for a 12-hour shift. A reading of -999 means the sensor dropped out. The shift engineer suspects that deposits are building up on the tubes.

$$Q_h = \dot{m}_h c_{p,h} (T_{h,in} - T_{h,out}) \quad Q_c = \dot{m}_c c_{p,c} (T_{c,out} - T_{c,in}) \quad \Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)} \quad U = \frac{Q_h}{A \Delta T_{lm}}$$

with $\Delta T_1 = T_{h,in} - T_{c,out}$ and $\Delta T_2 = T_{h,out} - T_{c,in}$ (counter-current).

Given

```

1 | mh, cph = 2.0, 2.2      # oil:   kg/s, kJ/(kg.K)
2 | mc, cpc = 1.5, 4.18    # water: kg/s, kJ/(kg.K)
3 | Area = 20.0            # m^2
4 | # (hour, Th_in, Th_out, Tc_in, Tc_out) all in deg C
5 | log = [(1, 150.2, 53.3, 25.1, 93.1), (2, 149.8, 53.3, 24.8, 92.5),
6 |        (3, 150.5, 54.4, 25.3, 92.7), (4, 150.0, 54.4, 25.0, 92.1),
7 |        (5, 149.6, 55.1, 25.2, -999), (6, 150.3, 55.8, 24.9, 91.2),
8 |        (7, 150.1, 56.5, 25.0, 90.7), (8, 149.9, 57.7, 25.4, 90.1),
9 |        (9, 150.4, 58.4, 25.1, 84.2), (10, 150.0, 58.8, 24.7, 88.7),
10 |        (11, 149.7, 59.8, 25.0, 88.1), (12, 150.2, 61.0, 25.2, 87.8)]

```

- (a) Write `duty(m, cp, dT)`, returning kW, and `lmt(dT1, dT2)`, using `math.log`. **Explain** in one comment line why the LMTD, not the arithmetic mean, is used. [2]
- (b) Loop over `log` and unpack each tuple. Skip any row that contains -999, printing a warning with `continue`. For every other hour, **estimate** Q_h , Q_c , the energy-balance closure error $|Q_h - Q_c|/Q_h \times 100$, and U in kW/(m² K). Store each result as a tuple and print an aligned table. **Distinguish** (flag) any hour whose closure error is above 5%. [4]
- (c) Keep only the hours with closure $\leq 5\%$. Using slicing, **compare** the mean U of the first three and last three of those hours, and report the % drop. Use `min(..., key=...)` to **identify** the hour with the lowest U . Take hour 1's U as the clean value, and find the first hour in which U falls below 80% of it. [2]
- (d) **Interpret**: (i) is the exchanger fouling? Quote the trend in U . (ii) Which hour did you flag, and why should it be left out of the U trend rather than “corrected”? (iii) When would you recommend taking the exchanger offline for cleaning? [2]

— End of Assignment —